Descriptive Logic V: Following a Rule

Philosophical Investigations §§134–242, in Phient's terms

Phiano Project · `bin/language_game` (`ask`, `follow`, `graded`, `compare`) · `bin/code_reader` · 2026-10-01

> In Phient's world, a word (w) is nothing but what it relates to, under what conditions (c). The Phical paradigm, X ⇒ Y, drives and composes Phient's Phasor Language Models: Phinet, Phiano, PhiTon. Phical stores no point: a word is 32–64 relations, each a complex phasor on a closed circle with a count; meaning is a side effect of pairs of words moving under a condition.

Chapters I–IV played the builders' game with teachers we wrote, then read Phi-4's own relations as rough ground. This chapter puts Phi-4 *in* the game as the other player. It is told the set-up (the stones, the letters used as numerals, the places, the form of an answer) and never the rules. Then it is asked, call by call, what it does.

§§134–242 ask what it is to follow a rule. When does a continuation count as *going on in the same way*? Where does understanding show itself? And why can no interpretation fix a rule by itself? In our project these questions have numbers. Phi-4 was asked the same 400 calls in three settings:

1. **sampled:** Ollama's default sampling;
2. **temperature 0:** always the likeliest answer;
3. **temperature 0 with the roles stated:** the scene also says who fetches and who only takes note.

Students then learned to go on as Phi-4 does.

As before, Wittgenstein is paraphrased and cited by remark, never quoted.

---

1. Four more definitions

- **D17 · Practice.** A teacher's answers form a *practice* when the same call gets the same answer. Its measure is *steadiness*: the share of pairs of answers to a repeated call that are identical (§§198–199).
- **D18 · Going on differently.** A divergence from our rule is *going on differently*, not error, when it follows the words of the call: it can be

predicted from them, and it repeats. Divergence that follows nothing is noise (§185).

- **D19 · Reading, not interpreting.** A teacher's notation is read in the game's terms only where the call gives no word for the aspect (`-` for no colour is the game's `any`). Every further reading would be one more interpretation of a rule by a rule (§§198, 201).
- **D20 · Agreement in judgments.** Teacher and game agree in judgments on an aspect when, read as in D19, the teacher's answers are the game's in at least 95% of calls. That agreement is not given by any definition the prompt contains (§242).

2. The data

Each cell is one run of 400 calls (`data/builders/`), read by `follow` and compared by `compare`:

run	replies read	exact (game's terms)	act	stone	colour	count	place	steadiness (D17)
sampled	369	50.7%	63.1%	94.9%	96.2%	89.7%	89.7%	35.8% (95 of 265)
temperature 0	393	66.4%	74.3%	98.2%	100%	94.1%	96.4%	88.9% (295 of 332)
temperature 0 + roles	380	**88.2%**	**99.5%**	97.4%	**100%**	93.7%	**97.1%**	**100%** (276 of 276)

Students learned Phi-4's practice from four fifths of the calls and predicted its answers on the fifth (folds by call, so a repeated call is never on both sides). The exact match:

student of Phi-4	sampled	temperature 0	t0 + roles
our rules, as a predictor	50.7%	66.4%	88.2%
bag of words + softmax	**54.2%**	**70.5%**	**89.7%**
counts, D2 (τ = 0.5)	46.6%	60.3%	85.3%
counts, τ per aspect, chosen in training (docs/111)	49.3%	65.1%	89.5%
counts, two-speed posts (docs/113)	47.2%	70.2%	84.5%

3. The remarks, in Phient's terms

§§134–142 — The general form; what fits

The remarks. "This is how things are" looks like the general form of a proposition. But it is itself a sentence that belongs to our language, and its use is shown by how it is used (§§134–136). Whether a word *fits* a sentence is shown by the use, not by a picture that comes before the mind when the word is understood (§§138–141). Such pictures could suggest other uses. Normal cases make the use clear; abnormal cases make it unclear what to say (§142).

In Phient's terms. Phi-4 knows the words. It names the stone in 94.9–98.2% of calls in every run, and the colour in 96.2–100%. What fits in the slot is not the problem. The problem was the act. In the first two runs Phi-4 knew what *slab* means and still did not know what to *do* with it. The word fitted; the use did not.

§§143–155 — The series, and "now I can go on"

The remarks. A pupil is taught to write a series. At some point he can continue it himself, and we say he has understood the system (§§143–146). Understanding looks like a state from which the right use flows. But the criterion for it is the use itself, not an inner state (§§146–150). When someone says *now I can go on*, no process need be going on in him. The words are justified by the circumstances (§§151–155).

In Phient's terms. The code reader (docs/109–110) is a pupil of the series. Given 367,000 tokens of our Rust, it is asked to continue each file where the text stops. Its *understanding* is nothing but its continuations: 60.0% / 62.0% of next tokens exactly right on two splits of unseen files, against the 4-gram's 55.7% / 57.2%. There is no state in it apart from the counts and the weights. The weights say that what it adds to the 4-gram is the context four to eight tokens back. That is where its "going on" comes from.

§§156–184 — Reading, and being guided

The remarks. Reading machines are taken as an example. The criterion for whether someone *reads* is what they do, not an inner experience of deriving the words from the signs (§§156–171). Being guided, and the "because" of reading, is a family of cases, not one experience (§§172–178). *Now I can do it* is not a report of a state either (§§179–184).

In Phient's terms. Phi-4's replies are read by `builders::parse`. In the first run the parser was the bottleneck: 31 of 400 replies were unread, because of terminal escapes, markdown table rules and letter numerals. Whether Phi-4 *read* the call is shown only by the replies we can read. Once the parser was fixed, 393 and 380 of 400 replies were read in the later runs. The 7 and 20 left have no line in the form that was asked for.

§§185–190 — The pupil who goes on differently

The remarks. A pupil has learned to add 2, and is right up to 1000. Asked to go on past 1000, he writes 1004, 1008. Told he is wrong, he says he went on *in the same way*. He understood our order the way we would understand *add 2 up to 1000, 4 up to 2000*. No explanation we gave excluded that reading in advance (§185). What we meant was not a mental act that ran ahead through all the steps (§§186–188). How the steps are determined is shown by how we were trained and how we use the formula (§§189–190).

In Phient's terms. In the first two runs, Phi-4 was this pupil. Called to *bring*, it answered *record* in 85 orders (sampled) and 76 (temperature 0). It also answered *bring* to 25 reports. The divergence follows the words (D18). In temperature-0 orders that name a colour, Phi-4 never answers *record* (0 of 136). When the order names a slab or a pillar, it answers *record* about 45% of the time. When the call begins with *slab*, it answered *record* 9 times out of 9.

So this is not noise; it is going on differently. Phi-4 read *A calls out to you* as compatible with taking note, and the prompt never excluded that. The further explanation, *A tells you what to do and you do it; you do not take notes*, did exclude it. Divergence on the act fell from 101 calls to 2, and the act rose to 99.5%. As in §185, the first explanation was not wrong. It left open a way of going on that we did not mean, and the second explanation closed it.

§§191–197 — The rails laid in advance

The remarks. It can seem that the steps are already taken, in a queer way, when we grasp a rule at a stroke (§§191–197). The use seems present all at once. But that "all at once" is a picture, and its use is our ordinary use.

In Phient's terms. A Phient word has no rails in it. It is 32–64 relations with counts. Its future uses are not stored; they are recomposed each time from the relations that hold a post under the call (D2, D4). What looked like Phi-4's grasp of *slab* at a stroke did not contain the act.

§§198–202 — Signposts, custom, the paradox, and privately

The remarks. A rule stands like a signpost. Whether it leaves room for doubt depends on how it is used (§198). Following a rule is not something one person could do only once. It is a custom, a use, an institution (§199). Here is the paradox: no course of action could be fixed by a rule, because any course can be made to accord with it by some interpretation. The answer is that there is a way of grasping a rule that is not an interpretation, shown in what we call obeying it and going against it in actual cases (§201). Therefore following a rule is a practice, and one cannot follow a rule privately (§202).

In Phient's terms. D17 measures the custom. The sampled Phi-4 gave the same answer to the same call only 36% of the time: there was no practice to learn, and no student beat 54%. At temperature 0, steadiness rose to 89% and the best student to 70%. With the roles stated it reached 100%, and the best student 89.7%. A rule that is not kept twice cannot be taught.

D19 is our guard against the paradox. Phi-4 writes `-` where our game says `any`. We read `-` as `any` *only* where the call had no colour-word. That one reading takes exact match from 59.7% to 88.2% in the roles run. We did not let further readings rescue other divergences. With enough readings, every answer could be made to accord with our rules. The divergences are left standing, and the students learn them: on the roles run the best students (89.5–89.7%) predict Phi-4 *better* than our own rules do (88.2%).

§§203–216 — Obeying an order; training; the same

The remarks. Following a rule is like obeying an order. We are trained to it and react to it in a particular way (§206). Imagine people of an unknown language: the common behaviour of mankind is the system of reference by which we interpret it (§206). Explanation by examples is not second-best. The examples, with gestures and a *and so on*, are the explanation (§208). *The same* is not given by a table of identity that fits every case (§§215–216).

In Phient's terms. The students are trained on examples only: no rules, no prompt, just calls and what followed. The student that does best depends on the teacher:

- On the teacher of **tendencies** (temperature 0, no roles), the two-speed posts did best among the counts students (70.2%, against bag of words at 70.5%). They let a word that has been seen often learn its own weight, so `slab` and `pillar` acquired their lean toward *record*.
- On the teacher of **rules** (roles stated), τ chosen per aspect in training tied with bag of words (89.5% against 89.7%), and the two-speed posts only added noise (84.5%).

What counts as *the same* step differs between the two teachers, and the students show it. No single setting fits both.

§§217–221 — Bedrock; obeying blindly

The remarks. If I have exhausted the justifications, I have reached bedrock: my spade is turned, and I say *this is simply what I do* (§217). The rails invisibly laid to infinity are only a picture (§218). When I obey a rule I do not choose. I obey it blindly (§219).

In Phient's terms. For a Phient student, the reasons give out at the counts. *Why bring?* Because `slab` holds a post under the act and moves it toward *bring*. *Why that post?* Because its counts move the answer by more than τ. *Why that τ?* Because, on these calls, that τ answered best in the training folds. Below that there is nothing but the told episodes.

Phi-4 at temperature 0 with the roles stated obeys blindly in the measurable sense: 276 of 276 repeated calls get identical answers.

§§222–237 — Agreement and rule are cousins

The remarks. The word *agreement* and the word *rule* are related, cousins (§224). To teach someone the use of one is to teach the use of the other. What *the same* means is learned with the rule (§§225–227). The rule can seem to tell us what to do as if from outside, but we follow it as a matter of course (§§228–237).

In Phient's terms. Agreement and rule are measured by the same numbers here. *Phi-4 follows our rule* is the first row of the student table, and *a student follows Phi-4's practice* is every other row. On the roles run, our rules agree with Phi-4 at 88.2%, and Phi-4's practice, learned by a student, agrees with Phi-4 at 89.7%. The residue is 11.8% divergence from our rules. Most of it is notation:

- *i* counted as 10 (13 calls);
- the colour written into the stone, as in `yellow block` (10 calls);
- a place `here` where the rule has none (11 calls).

§§238–242 — Agreement in judgments, and in form of life

The remarks. For the rule to seem to produce all its consequences in advance, they must be a matter of course to me (§238). Mathematicians do not come to blows over whether a rule was obeyed (§240). Whether people agree is not decided by their opinions. They agree in the language they use, which is agreement in form of life, not in opinions (§241). Communication by language needs agreement in definitions and also, odd as it may sound, agreement in judgments (§242).

In Phient's terms. D20 sorts the aspects:

aspect	sampled	t0	t0 + roles	D20 (≥ 95%)
stone (*what*)	94.9%	98.2%	97.4%	agrees from t0
colour	96.2%	100%	100%	agrees in every run
place	89.7%	96.4%	97.1%	agrees from t0
count	89.7%	94.1%	93.7%	not yet: `i` read as 10
act	63.1%	74.3%	99.5%	agrees only with the roles stated

The prompt gives definitions: the stones, the colours, the letters as numerals. Agreement in judgments on stone, colour and place came with those definitions. Agreement on the act did not. It needed more of the form of life: who fetches, who takes note. The prompt's definitions could not supply that, and Phi-4's training had not.

The count shows the other side of §242. The prompt *defines i* as 9, and Phi-4 still judges *i* as 10 in 13 calls. Definitions were given and agreed to, but the judgments did not follow from them.

4. What Chapter V adds

- **Practice can be measured.** Steadiness went 36% → 89% → 100%, and what a student could learn followed it: 54% → 70% → 90%.
- **Going on differently is predictable.** Phi-4's *record* for *bring* followed the stone named and the presence of a colour-word. One more sentence of explanation, as in §185, took it from 101 calls to 2.
- **The student learns the practice, not our rules.** On the roles run the best students predict Phi-4 better than our rules do.
- **Which student is best depends on the teacher** (D16 again): two-speed posts for a teacher of tendencies, τ per aspect for a teacher of rules. Counts students keep one thing bag of words cannot do: learn a word from one pointing (Chapter I, sepia: 100% against 0%).

5. Postscript: Phi-4 taught by pointing (§§28–30, docs/116)

Phi-4 was taught `sepia` by one or three pointings, with or without the word *colour*. It was then given 50 orders that use the word, at temperature 0, with the roles stated:

teaching	Phi-4: colour = `sepia`	Phi-4: act	Phi-4: exact	counts student: exact	student + Phi-4's reading
1 pointing	64.9%	54.1%	29.7%	32%	100%
3 pointings	90.0%	48.0%	34.0%	32%	100%
1 pointing + *colour*	63.6%	59.1%	27.3%	100%	100%
3 pointings + *colour*	90.0%	24.0%	16.0%	100%	100%

Phi-4 learns the word: three pointings put it in the colour slot in 90% of orders. But being taught moves it out of the game. Its act falls from 99.5% to 24–59%, and with three pointings and *colour*, 38 of 50 orders are answered *record*. The scene of the teaching leaks into the scene of the order: §29's question of which game a word is placed in, answered wrongly.

Our counts student keeps the pointing in its own game (D3). It does not need Phi-4's skill, only its reading. When Phi-4 takes the word as a colour in most orders, the student is told so and keeps everything else for itself. Exact is then 100% in all four teachings. This is the division of labour for the 4% model: the word's aspect from the large model, the game from the relation model.

6. Next

Next, the student learns from Phi-4's practice and from its night relations together. That is the first step of distilling Phi-4 into a relation model at the scale of the size knob.

Code: `phiano/src/bin/language_game/` (`teacher.rs`, `follow.rs`, `graded.rs`, `weighted.rs`, `wordposts.rs`, `compare.rs`), `phiecs/src/quest/builders.rs`, `phiano/src/bin/code_reader/`. *Docs:* 107, 109–115. *Data:* `data/builders/phi4_seed7*`, `phi4_t0_seed7*`, `phi4_t0_seed7_roles*`.

Run: `scripts\phi4_builders.ps1` (add `-Sampled` or `-Roles`), then `language_game compare data\builders\runs.md data\builders\phi4*.tsv`.